

Minimal Notes

Complex Analysis

ALBERT REHAK

2024/25

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WEEK 1 - COMPLEX ARITHMETIC AND CONTINUITY

Useful Formulas 1.

In the following $z, z_1, z_2 \in \mathbb{C}$ and $x, y \in \mathbb{R}$.

- $z\bar{z} = |z|^2$
- $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$
- $\operatorname{Re}(z) = \frac{z + \bar{z}}{2}$
- $\operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$
- $e^{x+iy} = e^x \cos y + ie^x \sin y$
- $|e^{x+iy}| = e^x$

Proposition 10. $z_n = x_n + iy_n \rightarrow z = x + iy \iff x_n \rightarrow x \text{ and } y_n \rightarrow y$.

Ex1 p5,6

Algebra of Limits

Definition 1 (Continuity using $\varepsilon - \delta$). $f(z)$ is continuous at z_0 if for every $\varepsilon > 0$ there is some $\delta > 0$ so that

$$\text{for any } z \text{ with } |z - z_0| < \delta \implies |f(z) - f(z_0)| < \varepsilon.$$

Algebra of continuous functions

Proposition (Continuity by components). $f(x + iy) = u(x, y) + iv(x, y)$ is continuous $\iff u$ and v are continuous.

WEEK 2 - DIFFERENTIATION

Definition 2 (Definition of a derivative). We have three equivalent definitions of the derivative:

- $f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}$.
- $f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$.
- $f(z_0 + h) = f(z_0) + hf'(z_0) + \text{error}(h)$ with $\text{error}(h)/h \rightarrow 0$ as $h \rightarrow 0$.

Algebra of derivatives including all types of chain rules

Proposition (Cauchy-Riemann equations). The following are equivalent necessary conditions for complex differentiability of $f(x + iy) = u(x, y) + iv(x, y)$:

- $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$.
- $\frac{\partial f}{\partial \bar{z}} := \frac{1}{2} \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \right) = 0$. These CR equations are often much quicker to check if you can write the function in terms of z and $\bar{z}$.

To get sufficient conditions for differentiability we (additionally to CR) need that u_x, v_x are continuous.

Example 1: When checking using the $\bar{z}$ formulation of CR equations one can think of z and $\bar{z}$ as two distinct variables. In this view the equation $\frac{\partial f}{\partial \bar{z}}$ becomes standard partial differentiation. E.g.

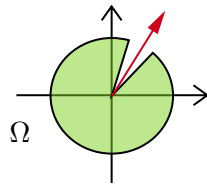
- $f(z) = |z|^2$ is not holomorphic because $\partial_{\bar{z}}(|z|^2) = \partial_{\bar{z}}(z\bar{z}) \neq 0$.
- Question: Is $f(z) = \operatorname{Im}(z)$ holomorphic?

Useful Formulas 2.

- $f'(z) = \frac{\partial f}{\partial z}$
- $\text{Log } z := \log |z| + i \arg(z) + 2\pi ki$
- $z^\alpha := e^{\alpha \text{Log } z}$
- $(\log z)' = \frac{1}{z}$

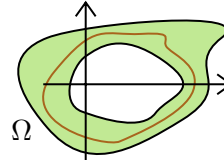
Note that $\arg(z)$ and $\log(z)$ ¹ are only well-defined on domains which do not contain a loop that winds around 0 a non-zero number of times. So for these to be well-defined it is sufficient that some infinite ray centered at 0 is excluded from the domain. Conversely, any domain which contains a full circle of non-zero radius centered at 0 is bad.

No loop in Ω intersects the red ray



GOOD DOMAIN

The red loop in Ω winds around zero



BAD DOMAIN

Why is the second domain bad? This is because there is a loop in Ω that winds around zero. If we take any point $z_0 = re^{i\theta}$ on the loop, then after making a full rotation around it we get additional 2π in the argument of z_0 . So $\arg(z_0) = \theta$ but also $\theta + 2\pi$. This would make $\arg(z)$ multivalued.

Proposition. The real and imaginary parts of a holomorphic function are harmonic (i.e. satisfy $u_{xx} + u_{yy} = 0 = v_{xx} + v_{yy}$). Ex2 p3,4

WEEK 3 - POWER SERIES

SERIES OF COMPLEX NUMBERS $\sum a_n$

Theorem 40 (Comparison test).

Proposition (Absolute convergence implies convergence).

Theorem 42 (Convergence tests).

- (Cauchy's root test) If $\limsup_{n \rightarrow \infty} |a_n|^{1/n} < 1$ then $\sum a_n$ converges absolutely. If > 1 , it diverges.
- (Ratio test) If $\lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|}$ exists and < 1 then $\sum a_n$ converges absolutely. If > 1 , it diverges.

Note that the root test is more powerful because the limit always exists. The tests are inconclusive if the limit does not exist or when it equals 1.

SERIES OF COMPLEX FUNCTIONS $\sum f_n(z)$

Theorem 46 (Weierstrass M-test). If for all z, n we have $|f_n(z)| \leq M_n$ and $\sum M_n < \infty$ then $\sum f_n(z)$ is uniformly convergent.

¹The problem of well-definedness of $\text{Log}(z)$ comes from $\arg(z)$ in its definition.

Idea (Radius of convergence). *The largest set on which a power series $\sum a_n(z - z_0)^n$ can converge is a ball $B_r(z_0) = \{z : |z - z_0| < r\}$ ². In particular, balls are the only admissible domain on which a series can converge. This is with the slight exception of the boundary of the ball where convergence has to be checked separately.*

To find the largest possible radius R of a ball on which a series converges we use either of the Ex4 p2 following:

- $R = (\limsup_{n \rightarrow \infty} |a_n|^{1/n})^{-1}$.
- If $\lim \frac{|a_n|}{|a_{n+1}|}$ exists then it equals R .

Example 2: If $\sum a_n(z - z_0)^n$ converges for some z with $|z - z_0| = r$ then the series automatically converges for all $z \in B_r(z_0)$.

Proposition (Absolute and Uniform convergences). *We have absolute convergence for any z such that $|z - z_0| < R$ and uniform convergence on any closed ball $\overline{B}_r(z_0)$ with $r < R$.*

Proposition. *Translating coefficients of a power series does not alter the radius of convergence.*

Theorem 55 (Term-by-term differentiation). *We can differentiate a power series within its radius of convergence term-by-term and the result has the same radius of convergence.*

$$\left(\sum a_n(z - z_0)^n \right)' = \sum (a_n(z - z_0)^n)' = \sum n a_n(z - z_0)^{n-1}.$$

From this it follows a power series is infinitely differentiable and by definition analytic. Its coefficients can be seen to be $a_n = \frac{f^{(n)}(z_0)}{n!}$ where $f = \sum a_n(z - z_0)^n$. It follows, we can integrate term-by-term. Ex4 p3

LAURENT SERIES

Definition 3 (Laurent Series). *We define a Laurent Series as*

$$\underbrace{\sum_{n=0}^{\infty} a_n(z - z_0)^n}_{\text{Taylor part}} + \underbrace{\sum_{n=-1}^{-\infty} a_n(z - z_0)^n}_{\text{Laurent part}}. \quad (1)$$

The *Taylor part* converges in a $B_{R^+}(z_0)$ with radius of convergence R^+ . The *Laurent part* converges in the set $\overline{B_{R^-}(z_0)}^c$ and we also call R^- the radius of convergence. In **Theorem 107** we will see that on the intersection of the two sets $z \in B_{R^+}(z_0) \cap \overline{B_{R^-}(z_0)}^c$ the expansion (1) is equal to f holomorphic on the intersection (with coefficients determined by f).

Example 3: We know that for any $|z| < |a|$ we have the geometric series expansion

$$\frac{1}{a - z} = \frac{1}{a(1 - \frac{z}{a})} = \frac{1}{a} \left(1 + \frac{z}{a} + \frac{z^2}{a^2} + \dots \right).$$

We will refer to this as the *Taylor type* expansion of $\frac{1}{a-z}$. Using this we observe

$$\frac{1}{a - z} = \frac{-1}{z(1 - \frac{a}{z})} = \frac{-1}{z} \left(1 + \frac{a}{z} + \frac{a^2}{z^2} + \dots \right) \text{ is valid for } \left| \frac{a}{z} \right| < 1 \iff |z| > |a|.$$

This is going to be called the *Laurent type* expansion of $\frac{1}{a-z}$.

Tips:

- Use partial fractions.
- Ask yourself whether $\frac{1}{z-a}$ should be expanded in the Taylor or Laurent type. This is determined by choosing which one of the $|z/a|$ (Taylor) or $|a/z|$ (Laurent) is smaller than 1. To do this one needs to use the domain on which the Laurent series is to be determined.

WEEK 4 - COMPLEX INTEGRATION I

Definition 4 (Contour Integral). Suppose that $\gamma : [a, b] \rightarrow \mathbb{C}$ is a (piecewise) smooth path. Then we define

$$\int_{\gamma} f(z) dz := \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

This definition is independent of parametrization of the image $\gamma([a, b])$.

Theorem 68 (FTC for contour integrals). Suppose F is differentiable on the image of a smooth curve $\gamma : [a, b] \rightarrow \mathbb{C}$ and F' is continuous here. Then

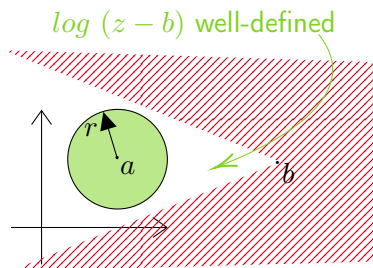
$$\int_{\gamma} F'(z) dz = F(\gamma(b)) - F(\gamma(a)).$$

Namely, if γ is closed then the integral equals 0. Similarly, if f has an anti-derivative then $\int_{\gamma} f(z) dz = 0$ for any closed γ .

Idea (FTC). The key takeaway of FTC for contour integrals is that $\int F'(z) dz$ is independent of the path. It only depends on the endpoints of the path! In particular $\int_{[z_0, z_1]} F'(z) dz = F(z_1) - F(z_0)$. Ex5 p3,5

Example 4: For $n \neq -1$ because z^n has an anti-derivative, we have $\int_{\gamma_1} z^n dz = \int_{\gamma_2} z^n dz$ for any smooth curves γ_1 and γ_2 with the same endpoints.

Example 5: Note that if $b \in B_r^c(a)$ we can define $\log(z - b)$ on some infinite sector which includes $B_r(a)$ and so get a primitive function $\log(z - b)$ of $1/(z - b)$. Ex5 p6



By FTC we have $\int_{\partial B_r(a)} 1/(z - b) dz = 0$. Here $\partial B_r(a)$ denotes the path $\gamma(t) = a + re^{it}$ with $t \in [0, 2\pi]$.

Useful Formulas 3. Let $\gamma(t) = a + re^{it}$ for $t \in [0, 2\pi]$ and $\eta : [a, b] \rightarrow \mathbb{C}$ any smooth path. Then Ex5 p1,8

²Notation and nomenclature in lectures was a disc $D(z_0, r)$.

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- | | |
|--|--|
| • $\int_{\gamma} (z - b)^n = 0$ for any $b \in \mathbb{C}$ and $n \neq -1$. | • $ \int_a^b \eta(t) dt \leq \int_a^b \eta(t) dt$. |
| • $\int_{\gamma} \frac{1}{z-b} = 2\pi i$ if $b \in B_r(a)$. | • $ \int_{\eta} f(z) dz \leq \text{length}(\eta) \sup_{z \in \eta([a,b])} f(z) $. |
| • $\int_{\gamma} \frac{1}{z-b} = 0$ if $b \in B_r^c(a)$. | • $\text{length}(\eta) = \int_a^b \eta'(t) dt$. |

We will see later³ that the formulas in the first column apply to any closed curve γ where the condition $b \in B_r(a)$ is replaced by b being in the interior region defined by the curve. This applies when γ winds around only once. If it winds around k times the result (for the second formula in the first column) is $2\pi ki$.

The last observation (coupled with the Homotopy Cauchy theorem) leads to the following definition.

Definition 5 (Winding number). For closed path γ with c not on the path we define

$$n(\gamma; c) = \frac{1}{2\pi i} \int_{\gamma} \frac{1}{z - c} dz.$$

Proposition (Properties of the Winding number).

- | | |
|---|--|
| • $n(\gamma, c) \in \mathbb{Z}$. | closed. |
| • $n(\gamma, c) = n(\gamma, d)$ if c, d belong to the same connected component of $\mathbb{C} \setminus \gamma$. | • $n(\gamma_0, c) = n(\gamma_1, c)$ if $\gamma_0 \sim_{\mathbb{C} \setminus \{c\}} \gamma_1$. |
| • $n(\gamma, c) = 0$ if c lies 'outside' the region en- | • $n(\gamma, c)$ is independent of parametrization. |

WEEK 5 - COMPLEX INTEGRATION II

Theorem 79 and 83. Let $\Omega \subset^{op} \mathbb{C}$ and $*$ -shaped with central point z_0 . If a continuous f satisfies

$$\int_{\partial \Delta} f(z) dz = 0 \quad \text{for any triangle } \Delta \subset \Omega \text{ with vertex at } z_0,$$

then f has an anti-derivative F on Ω . By FTC we obtain the Cauchy theorem on $*$ -shaped domains

$$\int_{\gamma} f(z) dz = 0 \quad \text{for any } \gamma \subset \Omega.$$

We will later see (Morera's theorem) that the second conclusion of the above theorem can be strengthened (at the cost of possibly losing an anti-derivative) to any open set Ω not only a $*$ -shaped one.

$$\int_{\partial \Delta} f(z) dz = 0 \quad \text{for any triangle } \Delta \subset \Omega \implies f \text{ holomorphic.}$$

The following theorem can be viewed as a converse of this implication.

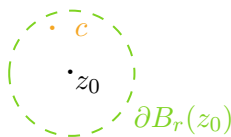
Theorem 81 (Cauchy-Goursat). Let $\Omega \subset^{op} \mathbb{C}$ and f holomorphic in Ω . Then

$$\int_{\partial \Delta} f(z) dz = 0 \quad \text{for any triangle } \Delta \subset \Omega.$$

The conclusion remains unchanged if we only assume f is holomorphic, bounded on $\Omega \setminus \{c\}$ for some $c \in \Omega$.

³As a consequence of Cauchy's theorem.

The following remarkable theorem says that it is sufficient to know the values of a holomorphic on the boundary of a ball $\partial B_r(z_0)$ to know the value the function attains on any point c on the inside of the ball.



Theorem 86 (Cauchy's Integral formula on a disc). *Suppose $f \in H(B_R(z_0))$ then for $0 < r < R$ we have*

$$f(c) = \frac{1}{2\pi i} \int_{\partial B_r(z_0)} \frac{f(z)}{z - c} dz \quad \text{for every } c \in B_r(z_0). \quad (2)$$

By choosing $c = z_0$ we obtain the Mean Value Property

Ex6 p8

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt.$$

WEEK 6 - SERIES EXPANSIONS OF HOLOMORPHIC FUNCTIONS

Theorem 87 (Any holomorphic function can be expanded as a power series in a ball). *Suppose that $f \in H(B_R(z_0))$ then*

$$f(z) = \sum_0^{\infty} a_n (z - z_0)^n \quad \text{for all } z \in B(z_0, R).$$

Integrating by parts (2), we can see that for $0 < r < R$ the coefficients a_n are determined by

$$a_n = \frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i} \int_{\partial B_r(z_0)} \frac{f(w)}{(w - z_0)^{n+1}} dw.$$

Theorem 89 (Liouville's theorem). *A bounded holomorphic function on the whole complex plane $\mathbb{C}$ is constant.*

An important corollary of Liouville's theorem is the fundamental theorem of algebra which says that a polynomial of degree n has exactly n (possibly repeated) complex roots.

Theorem 93 (Maximum Modulus Principle). *Suppose that f is holomorphic in an open connected $\Omega \subset \mathbb{C}$. Then any such non-constant function f cannot have **local** maxima in Ω . In particular if f is continuous on $\overline{\Omega}$ then for any $z \in \Omega$*

$$|f(z)| \leq \sup_{w \in \partial\Omega} |f(w)|.$$

Intuitively: holomorphic functions are largest on the boundary of a set.

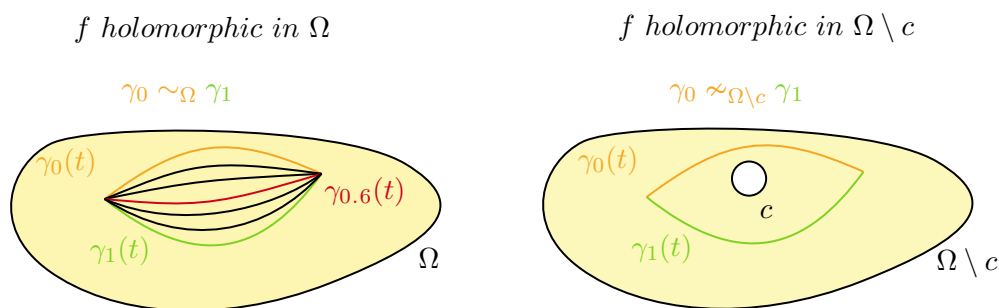
WEEK 7 - HOMOTOPY CAUCHY THEOREM

Theorem 95 and 96 (Cauchy Homotopy). *Let $\Omega^{op} \subset \mathbb{C}$ and suppose $f \in H(\Omega)$. Then for any paths $\gamma_1 \sim \gamma_2$ **homotopic in Ω** with the same endpoints we have*

$$\int_{\gamma_1} f = \int_{\gamma_2} f.$$

Equivalently, if a closed path is homotopic to any constant path $\gamma_{const.}(t) = c$, then from above $\int_{\gamma} f = \int_{\gamma_{const.}} f := \int_a^b f(\gamma_{const.}(t))\gamma'_{const.}(t)dt = 0$ because $\gamma'_{const.}(t) = 0$.

The following illustration shows when two paths are not homotopic. We think of c as a **hole** in Ω . Think of γ_0 and γ_1 as infinitely stretchable and contractible strings. It is then clear that if there are no holes in Ω then we can reshape γ_0 to γ_1 so $\gamma_0 \sim_{\Omega} \gamma_1$ ⁴. On the other hand, if there is a hole (even if c is a single point!) then $\gamma_0 \not\sim_{\Omega \setminus c} \gamma_1$ because there is no way to reshape the string γ_0 into γ_1 .



Example 6 (Singularities are holes): How do such holes come to be? Recall homotopy is highly dependent on the shape of the region where we are reshaping the paths. The region of reshaping is the region where f is holomorphic. For example the function $f(z) = \frac{1}{z-c}$ is holomorphic everywhere except at c . The point c is often referred to as a singularity. Because f is not holomorphic at c we can think of this singularity as a hole.

Theorem 97 (Cauchy Integral Formula via Homotopy). *Let $\Omega^{op} \subset \mathbb{C}$ and suppose $f \in H(\Omega)$. Then for any paths $\gamma \sim \gamma_{const.}$ and any $c \in \Omega \setminus \gamma((a, b))$ ⁵ then*

$$n(\gamma, c)f(c) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z-c} dz.$$

Namely, if c lies 'outside' the region enclosed by γ the integral is zero. In fact, we have

$$n(\gamma, c)f^{(n)}(c) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-c)^{n+1}} dz.$$

Ex6
p2,3,4
Ex7 p5,6

The following theorem is perhaps the most important theorem of complex analysis.

Theorem 1. *The following are equivalent:*

- $f \in H(\Omega)$.

⁴Homotopy does however differentiate between the number of times a loop winds around a hole. This agrees with thinking of the paths as strings. So if γ_0 wind once around a hole it is **not** homotopic to γ_1 if that winds twice around the hole.

⁵ c can be anywhere in Ω it just cannot lie on the path γ .

- f has a primitive function in Ω .
- $\int_{\Delta} f(z) dz = 0$ for every closed triangle $\Delta : [a, b] \rightarrow \Omega$.
- f equals its power series expansion. Namely, f is infinitely differentiable.

The following theorem says that a holomorphic function on an annulus is equal to its Laurent series.

Theorem 107 (Laurent Series on an annulus). Suppose $f \in H(A(z_0, R_1, R_2))$. Then

Ex8 p1

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n \quad \text{for all } z \in A(z_0, R_1, R_2).$$

For $R_1 < r < R_2$ and any $n \in \mathbb{Z}$ the coefficients are

$$a_n = \frac{1}{2\pi i} \int_{\partial B_r(z_0)} \frac{f(w)}{(w - z_0)^{n+1}}.$$

Example 7 (Finding Laurent series of functions): Similarly to Ex6 q5, we can multiply Laurent series. In particular, we can use the well-known Taylor series of standard functions to find the Laurent expansion of such functions multiplied by a polynomial. For example to find the Laurent expansion of $f(z) = e^z/z$ around $z = 0$ we can use the Taylor series of $e^z = 1 + z + z^2/2! + \dots$ which we know converges for every z and just multiply it by $1/z$ which makes sense for any $|z| > 0$. We obtain $e^z/z = 1/z + 1 + z/2! + \dots$ for $|z| > 0$.

WEEK 8 - SINGULARITIES AND RESIDUE THEOREM

SINGULARITIES

Definition 6 (Singularities). We differentiate singularities depending on the lowest power (non-zero coefficient) term in the Laurent expansion of f . Ex10 p2

- If $f(z) = a_0 + a_1(z - z_0) + \dots + a_n(z - z_0)^n + \dots$ on $B_R(z_0) \setminus \{z_0\}$ then z_0 is called a removable singularity.
- If $f(z) = \frac{a_{-m}}{(z - z_0)^m} + \dots + a_0 + a_1(z - z_0) + \dots + a_n(z - z_0)^n + \dots$ on $B_R(z_0) \setminus \{z_0\}$ then z_0 is called a pole of order m .
- If there is no smallest power in the Laurent expansion of f at z_0 i.e. we can find arbitrarily large negative non-zero coefficients, then z_0 is called essential singularity.

We note f has a pole of order m at z_0 iff $1/f$ has zero of order m at z_0 .

Example 8: To give examples of singularities it suffices to take a function with a well-known Taylor expansion and alter it so that it satisfies the required definition. For example, we know $e^z = 1 + z + z^2/2! + \dots$. To get a removable singularity at $z = 0$ we can take $\frac{e^z - 1}{z}$, pole of order m can be achieved using the function e^z/z^m , and essential singularity at $z = 0$ can be, for example, $e^{1/z}$ because the negative powers never terminate.

It is useful to keep in mind equivalent characterizations of singularities so that we can decide what kind of singularity a function has at the point z_0 without having to find the Laurent expansion which is laborious. For this purpose we have

Theorem 116, 117.

RESIDUES

Suppose for a moment that we can expand f in some $B_R(z_0) \setminus \{z_0\}$ as a Laurent series:

$$f(z) = a_0 + a_1(z - z_0)^1 + a_2(z - z_0)^2 + \dots \\ + \frac{a_{-1}}{(z - z_0)^1} + \frac{a_{-2}}{(z - z_0)^2} + \dots$$

We know we can both differentiate and integrate term-by-term inside the ball. So from Cauchy's theorem or **Useful Formulas 3** we see that all elements except $\frac{a_{-1}}{(z - z_0)^1}$ have zero integral over $\partial B_r(z_0)$. Using **Useful Formulas 3** again we obtain

$$\int_{\partial B_r(z_0)} f(z) dz = \int_{\partial B_r(z_0)} \frac{a_{-1}}{(z - z_0)^1} dz = a_{-1} 2\pi i.$$

The idea is that the only part of a holomorphic function in the region $B_R(z_0) \setminus \{z_0\}$ that has any effect on the result of the integral is the coefficient a_{-1} . This bears a special name.

Definition 7 (Residue).

$$\text{res}(f, z_0) := a_{-1} = \frac{1}{2\pi i} \int_{\partial B_r(z_0)} f(z) dz.$$

Importantly, we take the ball $B_r(z_0)$ so that no other singularity of f is inside.

Useful Formulas 4 (Formulas for the residue). • It appears that to find the residue of a function at z_0 we need to find the Laurent expansion at z_0 . However, in case f has pole of order m at z_0 we have a much easier formula. Ex8 p3

$$\text{res}(f, z_0) = \lim_{z \rightarrow z_0} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)].$$

In particular, if z_0 is a removable singularity then

$$\text{res}(f, z_0) = \lim_{z \rightarrow z_0} (z - a)f(z).$$

Also when f is of the form $\frac{\phi(z)}{(z - z_0)^m}$. Then $= \frac{\phi^{(m-1)}(z_0)}{(m-1)!}$ ⁶.

- If $f = \frac{\phi}{\psi}$ where ϕ and ψ are holomorphic in a neighborhood of z_0 then $\text{res}(f, z_0) = \frac{\phi(z_0)}{\psi'(z_0)}$.

Theorem 110 (Residue Theorem). Suppose that $f \in H(\Omega \setminus \{c_1, \dots, c_k\})$. If we can reshape γ to a constant path then Ex8 p2,
Ex9 p1

$$\int_{\gamma} f = 2\pi i \sum_{j=1}^k n(\gamma, c_j) \text{res}(f, c_j).$$

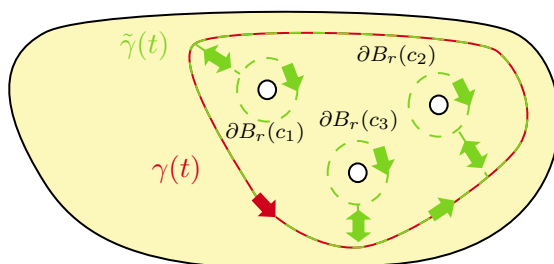
We provide a simple proof to demonstrate some of the theorems we have meet so far.

⁶The bracketed $(m-1)$ in the exponent means $m-1$ -st derivative not $m-1$ power.

Proof. Wlog let $k = 3$. Consider an alternative path $\tilde{\gamma}(t)$. The idea is that this alternative path avoids the singularities by encircling them in some small balls. Thus by Cauchy's theorem $\int_{\tilde{\gamma}} f = 0$. The paths connecting the balls to the green curve mimicking the original red curve have double arrows because we traverse them in both directions. We know that traversing back and forth gives a zero integral. This means the integral of the alternative path is just sum of the integrals over the balls and the original path:

$$0 = \int_{\tilde{\gamma}} f = \int_{\gamma} f - \int_{\partial B_r(c_1)} f - \int_{\partial B_r(c_2)} f - \int_{\partial B_r(c_3)} f$$

The minus signs appear because the balls are traversed in a clockwise direction whereas the original setting contains an anticlockwise traversing⁷. Of course from above, we notice that the three integrals are exactly the residues.



This proof is easily adapted to the case when γ winds multiple times around some of the singularities. This gives the general form including the winding numbers. $\square$

WEEK 9 - REAL INTEGRALS

One of the most practical uses of complex analytic tools is evaluation of real integrals. Real integrals can be thought of as path integrals over a path that lies strictly on the real axis. By complexifying the integrand and attaching a path (typically semicircular arc) to the real path so that we get a closed path in the complex plane, we can use many of the theorems (Residue, Cauchy) we have seen to help us evaluate the integral.

Theorem 113 (Real integral of polynomial fraction). *Suppose that $z_1, \dots, z_k$ are the roots of q .*

Ex9 p3,4

- If $\deg(q) \geq \deg(p) + 2$, then

$$\int_{-\infty}^{\infty} \frac{p(x)}{q(x)} dx = 2\pi i \sum_{\text{Im}(z_j) \geq 0} \text{res} \left(\frac{p(z)}{q(z)}, z_j \right).$$

That is, the sum is taken only over the roots which are in the upper half plane.

- If $\lambda > 0$ and $\deg(q) \geq \deg(p) + 1$, then

$$\int_{-\infty}^{\infty} e^{i\lambda x} \frac{p(x)}{q(x)} dx = 2\pi i \sum_{\text{Im}(z_j) \geq 0} \text{res} \left(e^{i\lambda z} \frac{p(z)}{q(z)}, z_j \right).$$

⁷The ball clockwise direction is necessary to assure that the green path avoids singularities. If it were anticlockwise, the singularities would be inside.

Tips:

- Whenever sin or cos appear in a real integral, it is often easier to compute instead the integral of e^{iz} and then take the real or imaginary part.
- If the first point fails, we can use the substitution $\theta = e^{iz}$ ($d\theta = \frac{dz}{iz}$) and turn integrals involving $\sin \theta$ or $\cos \theta$ into integrals in the z variable. Ex9 p1,2
- Computing residues is often easiest using **Useful Formulas 4**.

Example 9: Try computing $\int_{-\infty}^{\infty} \frac{x \cos x}{x^2 + a^2} dx$. Recipe: Suppose we want to compute over the upper arc. This means we have to identify the roots of the denominator which lie in the upper half plane. Now using **Useful Formulas 4** we compute the residue of p/q at these roots. Their sum is the result by **Theorem 113**.

WEEK 10 - ROUCHÉ AND ARGUMENT PRINCIPLE

Definition 8 (Meromorphic function). We say $f : \Omega \rightarrow \mathbb{C}$ is meromorphic if it is holomorphic in $\Omega \subset^{op} \mathbb{C}$ except possibly for poles.

Theorem 123 (Factorization of meromorphic functions). Say that f is meromorphic on Ω . Then we can factorize f as follows

$$f(z) = \frac{(z - a_1)^{n_1} \cdots (z - a_k)^{n_k}}{(z - b_1)^{m_1} \cdots (z - b_l)^{m_l}} h(z),$$

where a_j are zeros of f with multiplicities n_j , b_j are poles of f with multiplicities m_j and h is holomorphic with no zeros in Ω . Namely (by continuity) in a small ball $B_\varepsilon(a_j)$ for the zero a_j with multiplicity n_j we can write

$$f(z) = (z - a_j)^{n_j} \phi(z) \quad \text{where } \phi \text{ is holomorphic in } B_\varepsilon(a_j) \text{ with no other zeros.}$$

Similarly, for a pole at b_j with multiplicity m_j we can write

$$f(z) = \frac{\phi(z)}{(z - b_j)^{m_j}}.$$

Using the factorization above an easy computation near a_j yields

$$\frac{f'(z)}{f(z)} = \frac{n_j}{(z - a_j)} + \frac{\phi'(z)}{\phi(z)} \implies \operatorname{res} \left(\frac{f'}{f}, a_j \right) = n_j.$$

Similarly, for a pole b_j we obtain $\operatorname{res} \left(\frac{f'}{f}, b_j \right) = m_j$. By the residue theorem

$$\frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z)} dz = \sum_j n(\gamma, a_j) n_j - \sum_j n(\gamma, b_j) m_j.$$

This proves the so-called **Argument Principle** (Theorem 124). We think of the right side as $N - P$ with N the number of zeros (counting multiplicity) minus P the number of poles (counting multiplicity).

This following theorem appears on just about every complex analysis exam in the form: show that a polynomial has all roots in the annulus $A(r_1, r_2)$. It says that 'close' functions have the same number of zeros.

Theorem 127 (Rouché). *Let $\Omega \subset^{op} \mathbb{C}$ and $f, g \in H(\Omega)$. Suppose that γ is homotopic to some constant path. If* Ex10
p9,10

$$|g(z)| < |f(z)| \quad \text{for all } z \text{ on the path } \gamma,$$

then f and $f + g$ have the same number of zeros on the inside of γ .